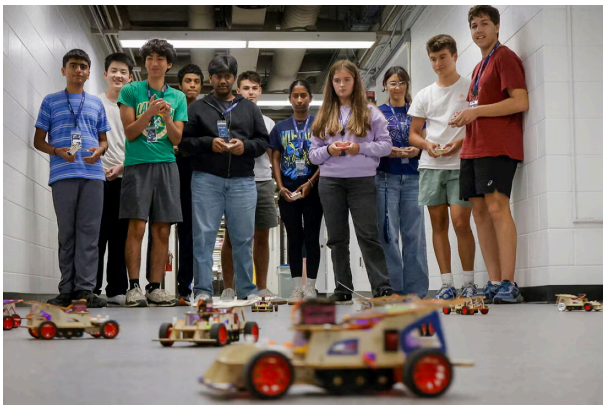
Summer camp gives high school students hands-on engineering experience

David Staudacher | Posted on **August 21, 2026**



UIC's inaugural Industrial, Mechanical, Prototyping, AI, and Computational Technologies (IMPACT) camp was a huge success, with both five-day sessions selling out quickly and more than 150 students attending.

"The program was created for ambitious high school students interested in learning more about design, manufacturing, and AI, and wanted to bridge the gap between theory and hands-on experiments said MIE Associate Professor Pranav Bhounsule[https://mie.uic.edu/profiles/bhounsule-pranav/], director of the Robotics and Motion Laboratory[https://pab47.github.io/].

The camp covered design engineering and reverse engineering, automotive engineering, applied AI and machine learning, fluid dynamics and hydraulics, and smart manufacturing.

"The students built a handheld electric fan from salvaged parts, a remote-operated rover with working steering, and an AI-assisted website, which included hands-on work with LLMs. They also worked with a robotic manipulator arm and applied Lean Six Sigma principles to design an automated candy-sorting machine," Bhounsule said.

The camp also introduced students to the UIC Engineering campus, which included tours of research labs and the UIC Makerspace. Each week was wrapped up with a closing ceremony where students received certificates of completion in front of instructors, family, and friends.

"We are very happy with the UIC IMPACT Camp experience. My daughter really enjoyed the opportunity to explore engineering in a hands-on and engaging way. She especially liked working on unique projects, solving problems, and using her creativity to come up with solutions. The teachers' expertise and knowledge had a very positive impact on the class and helped keep the students motivated and engaged," said Liana Madura, the parent of Giuliana Madura, a sophomore at Maine South High School. "She came home excited each day to talk about what she learned and the projects she worked on, which was great to see. Additionally, IMPACT allowed her to better understand the various engineering fields so that she could be more knowledgeable regarding potential areas to hone in on for her future engineering career path. We look forward to the opportunity to participate in this camp again next summer!"

In addition to the high school students, the UIC students enjoyed teaching the sessions and introducing engineering concepts and career options.

"Teaching is an absolute passion of mine, and we wanted to deliver a high-level experience that could challenge advanced senior students and be insightful," said MIE graduate student Paul Peretz, who co-directed the camp with Bhounsule and was also the main instructor. "I really enjoyed seeing students become increasingly curious throughout the week. It was a common theme every day where we encouraged the students to think outside the box and explore different ideas. Seeing everyone's lightbulbs turn on at one point or another is my favorite part of teaching."

Bhounsule and his team plan to offer the IMPACT program in 2027 [https://engineering.uic.edu/about/summer-special-programs/impact-summer-engineering-experience/] to high

school sophomores, juniors, and seniors and to expand it to eighth- and ninth-grade students. They also plan to add more five-day sessions to accommodate more students. Registration is slated to open in January.

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